

How long is an ED stay?

From a continuous clinical outcome to patient-disjoint classification and regression baselines

M2 teaches students to

- describe a right-skewed duration without hiding its tail
- compare groups without turning association into causation
- build and evaluate binary and continuous baselines

Carry the data contract forward



Row grain

one row represents one ED stay



Time boundary

triage information is separated from later events



Data quality

raw fields remain visible; imputed fields are explicit



Leakage

downstream outcomes and identifiers are excluded from predictors

M2 inherits the row, timing, and quality decisions from M1; it changes the label to ED length of stay.

What is ED length of stay?

ED LOS is the elapsed time from emergency-department arrival to ED departure.



Units matter: M2 measures the outcome in hours, not a diagnosis or a quality score.

Why ED LOS matters

Patient experience

time in the ED affects comfort, uncertainty, and access to the next step

Flow and capacity

duration is connected to rooms, staff, diagnostics, beds, and handoffs

Planning

estimates can support staffing and operational planning when uncertainty is visible

Measurement

the outcome exposes both clinical complexity and system workflow

Interpretation question: what part of LOS is clinical, and what part is operational?

Define the cohort before describing it

425,087

rows in the source table

425,081

valid non-negative LOS rows

62.02%

long-stay prevalence

53

zero-hour stays retained



Missing LOS

0 rows are not available for this outcome analysis.



Negative LOS

6 rows are impossible durations and are excluded.

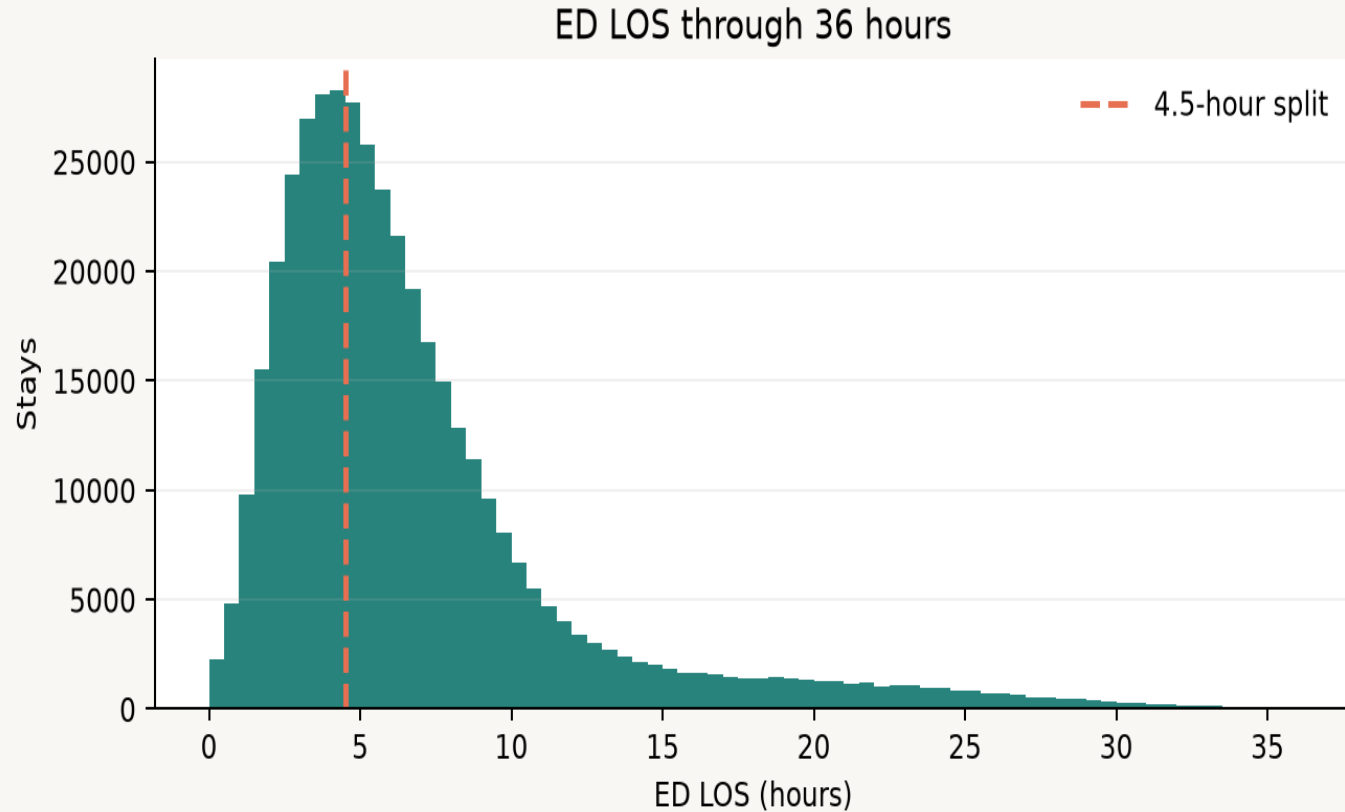


Valid extremes

34,342 valid rows exceed the 1.5 x IQR upper fence and remain in the cohort.

Quality rules change the analysis cohort, but the original table is not silently overwritten.

Continuous LOS: center, spread, and tail



Current cohort

5.47 h

median: middle stay

4.79 h

IQR: middle 50%

30.48 h

99th percentile: tail

The mean is 7.16 h; the maximum is 493.07 h.

The 4.5-hour split creates a second target

$\text{los_long_4_5h} = 1$ when ED LOS > 4.5 hours

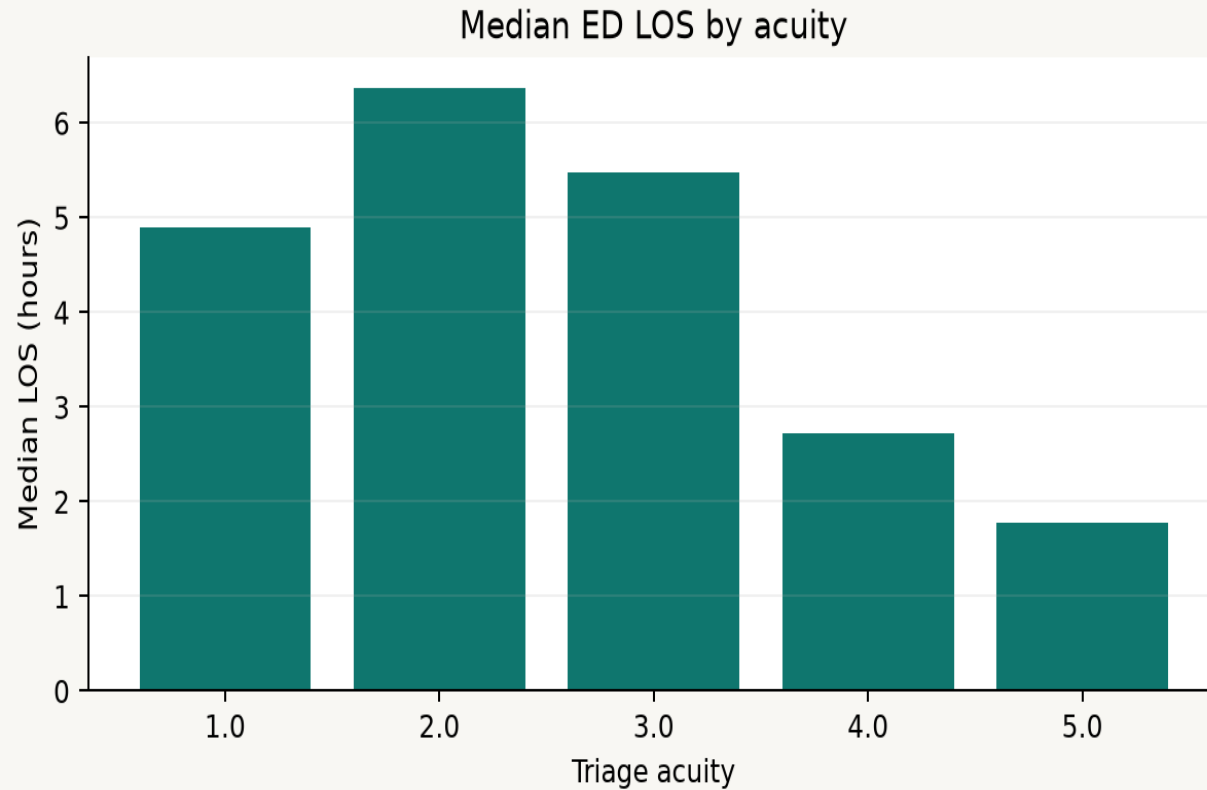


62.02%

of valid stays are above the threshold

Useful for an operational teaching question, but not a natural law. A 4.49-hour stay and a 4.51-hour stay are nearly identical.

Group comparisons generate questions



Read the pattern carefully

- Acuity 2 median: 6.37 hours.
- Acuity 4 median: 2.72 hours.
- Group sizes differ, so percentages need denominators.
- Association may reflect case mix and workflow, not causation.

M2 also compares age groups and arrival transport.

Start with the information boundary

Triage and context

age, gender, race, arrival
transport, imputed triage vitals,
acuity

Prior utilization

ED, hospital, and ICU counts
over 30, 90, and 365 days

Clinical flags

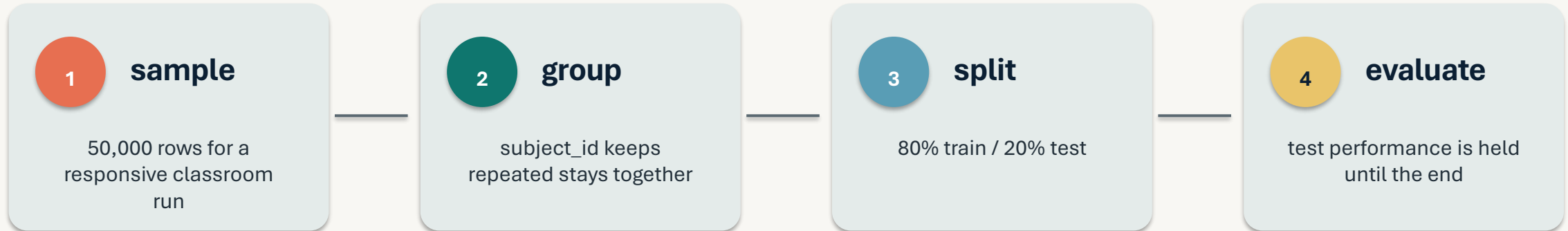
chief-complaint indicators and
CCI/ECI comorbidity indicators

Excluded from this triage-time baseline

identifiers | timestamps | disposition | ED LOS | outcomes | later ED measurements | medications

The model may be predictive only if its inputs could exist at the intended prediction time.

Train/test splitting: protect the patient boundary



40,009

training rows

9,991

test rows

32,360

train patients

8,090

test patients

Patient overlap in the split: 0

Classification models: different assumptions, same test set

	Prior dummy	predicts the training prevalence; a minimum reference	no patient information
	Logistic regression	linear decision boundary after preprocessing	transparent baseline
	Decision tree	recursive rules and interactions	interpretable splits
	Random forest	many trees averaged together	nonlinear ensemble
	Gradient boosting	sequential trees correct earlier errors	strong tabular baseline

Fair comparison means shared features, shared preprocessing, shared split, and the same held-out test set.

Classification metrics answer different questions

Sensitivity

of true long stays, how many were flagged?

Specificity

of shorter stays, how many were left unflagged?

PPV

of flagged stays, how many were actually long?

F1

a balance of precision and recall

AUROC

ranking quality across all thresholds

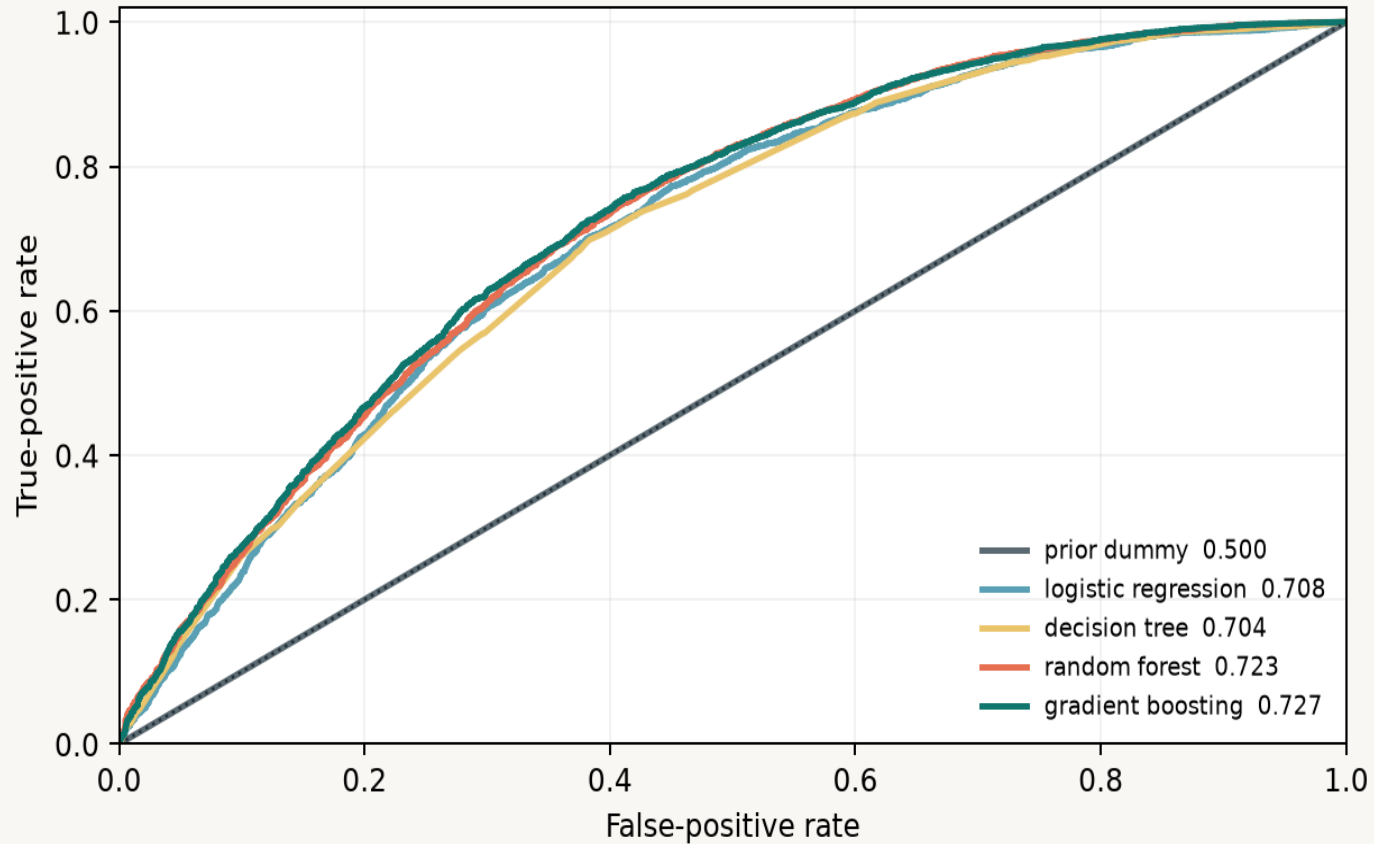
Average precision

precision-recall summary, useful with imbalance

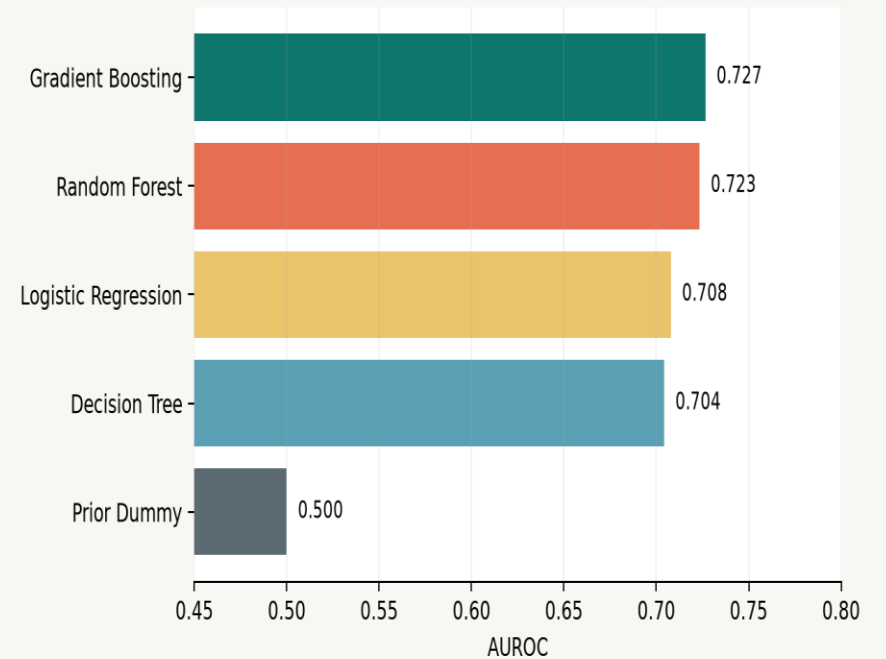
No single metric is “the performance.” The right metric depends on the action and the cost of errors.

AUROC: compare ranking across thresholds

ROC curves on the patient-disjoint test set



AUROC comparison



Highest AUROC: gradient boosting (0.727)

The threshold changes the error profile

The logistic model's ranking stays the same; the action changes when the cutoff changes.

Threshold	Sensitivity	Specificity	PPV	False positives
0.25	0.983	0.145	0.649	3,278
0.40	0.836	0.461	0.714	2,066
0.50	0.660	0.651	0.752	1,338
0.60	0.443	0.793	0.774	795
0.75	0.126	0.949	0.800	194

Operational reading

- Lower cutoff: more sensitivity, more alerts.
- Higher cutoff: fewer false positives, more missed long stays.
- Choose a cutoff with a real action and capacity constraint in mind.

Regression keeps the hours

Ridge regression predicts a continuous ED LOS rather than only a class.

Ridge

linear prediction with L2
regularization

shared preprocessing ->
transformed features ->
predicted hours

3.80 h

MAE: average absolute miss

6.52 h

RMSE: penalizes large misses

0.066

R2: variance-relative summary

**Read residuals too: a reasonable
average error can hide systematic
underprediction of the longest stays.**

A continuous prediction is more informative, but its error must still connect to a decision.

Binary and continuous targets support different decisions

Classification

- Question: will LOS exceed 4.5 hours?
- Output: probability or class label.
- Useful for alerts, routing, or thresholded actions.
- Cost: loses differences within each class.

Regression

- Question: how many hours is the stay?
- Output: expected continuous duration.
- Useful for planning and resource forecasting.
- Cost: errors are harder to translate into one action.

State the intended decision before choosing the target representation and metric.

What M2 exports and what students should inspect

**M2_los_summary.csv**center, spread, percentiles,
and tail of valid LOS**M2_los_by_acuity.csv**descriptive LOS and long-
stay rates by acuity**M2_los_by_age.csv**

age-group comparisons

**M2_los_by_arrival.csv**arrival-transport
comparisons**M2_classification_results.csv**combined metrics for all
classifiers**M2_threshold_results.csv**

logistic threshold trade-offs

**M2_regression_results.csv**

Ridge MAE, RMSE, and R2

The handoff is an argument: cohort -> target -> boundary -> split -> metric -> limitation.

The M2 mental model



A model score is only meaningful after the question, time boundary, cohort, and action are clear.

Next: carry the same discipline into the critical-outcome analysis.